

Noise-tailored two-qubit gates on a high-contrast featured dephasing model

Qasim Khan

June 13, 2026

This note is the companion to the June 11 report: there, the paper’s pipeline ran on the experimentally anchored Tarucha-class model and the noise-tailoring margin came out a reproducible but modest $1.5\text{--}1.7\times$. Here the same pipeline runs on a featured model — every structural ingredient still a measured type of spectral feature, but their placement and contrast chosen so that the value of the full spectral reconstruction is visible at a glance. This June 13 revision sharpens one ingredient: the top-window defect line is now a shared two-level fluctuator — common-mode between the qubits and, unlike the slow carrier, seated inside the decoupling passband. It changes no self-spectrum, so every average-fidelity result of the earlier featured run (the CZ ladder, both gates’ margins) is preserved to second order; what it adds is a strong, decoupling-robust correlation in $\text{Re } S_{1,2}$ that lets the idle gate carry the same two-qubit-necessity story the entangler already told. Three requirements imposed in the June 12 discussion still hold: the working infidelities must be realistic for today’s devices; the reconstruction must genuinely capture the model — every harmonic of all six spectra within its stated error bars; and the demonstration must articulate what the two-qubit part of the reconstruction buys. Throughout, the optimizer never sees ground truth, and every quoted “true” infidelity is evaluated exactly on the analytic model. All numbers June 13, 2026: the characterization is one 128k-shot reconstruction (everything downstream — designs, budgets, bands — is built on it), and the SPAM comparison is three 64k arms run on identical noise.

1 The noise model: what we assume, and why

Two correlated local fields, not one shared bath. The skeleton is unchanged from the anchored model: two partially correlated local electric fields e_A, e_B (shared fraction 0.8), the exchange coupling responding to their *difference*, $\zeta_{12} = a e_A - b e_B$ with $b \gtrsim a$, plus independent qubit-local processes. This is the minimal structure behind Yoneda’s measured $(+, +, -)$ coherence sign pattern, and it pins the signs and relative phases of all three cross-spectra rather than leaving them free. Five ingredients are then recomposed on this skeleton, each a measured feature type with its amplitude as the stated modeling choice; the one new to this revision is a *shared* fluctuator that carries the cross-spectra story below.

A slow, common-mode carrier holds the coherence budget. A slow bath (correlation time $\sim 400\tau$, in-band ω^{-2} tail) carries the entire $T_2^* = 3500\tau$ per qubit — the measured Ge-hole value ($17.6\mu\text{s}$ at the $\tau = 5\text{ ns}$ anchor), equivalently purified ^{28}Si at $\tau \approx 7\text{ ns}$. Its in-band tail is what defeats the low CDD orders. Structurally, the carrier is *common-mode* between the qubits at fraction $c_{\text{QS}} = 0.95$, with the remainder qubit-local: slow drifts of this class (global field, thermal, common charge motion) hit both qubits together, and inter-qubit coherences up to 0.7 are what Yoneda measured at low frequency. Two consequences are wired into the construction: the carrier appears in $\text{Re } S_{1,2}$ (and only there — a common-mode drift cancels in the difference coupling that drives ζ_{12} , so $S_{12,12}, S_{1,12}, S_{2,12}$ carry none of it), and its *sign* at low frequency is a measurable prediction with design consequences (Sec. 3).

The trap structure: a defect line with harmonics. On the qubit-local channels, a narrow defect line at $\omega_0\tau = 0.051$ with harmonics at $\{2, 3, 4\}\omega_0$ (cf. the ^{73}Ge Larmor line and its measured second harmonic surviving isotopic purification), plus one line at $\omega\tau = 0.372$ covering the top of the band reachable under the control-bandwidth constraint below. The placement is the engineered part: the CDD library’s passbands at $T_G = 320\tau$ fall on the first, second, and fourth harmonic for orders 3/4/5, and because the passbands scale as $1/T_G$ while the lines are fixed, *every doubling of the gate time slides the library down onto the next line*. This is why the margins in Sec. 4 grow with gate time. The harmonic family (ω_0 through $4\omega_0$) is strictly *local* defect structure — it appears on $S_{1,1}$ and $S_{2,2}$ only, and the inter-qubit coherence dips at those frequencies.

A shared fluctuator on the top-window line — the cross-spectra engine. The top-window line at $\omega\tau = 0.372$ is the exception, and the heart of this revision. It is not a local trap but a *shared* two-level fluctuator:

one defect coupling to both dots (a barrier charge trap or a shared phonon mode — the measured correlated-defect class), hence common-mode like the slow carrier. It carries the *same* power on each self-spectrum — the qubit $S_{1,1}/S_{2,2}$ totals, and with them every average-fidelity result of the previous report, are unchanged to second order — but it now also appears, real and positive, as a resonant peak in $\text{Re } S_{1,2}$. Two facts make it the operational core of the two-qubit story. Like the carrier it is common-mode, so it cancels in the difference coupling ζ_{12} and is invisible both to single-qubit QNS and, to first order, to average fidelity. But *unlike* the carrier its frequency sits *inside* the decoupling passband, not at DC: a ZZ -decoupling train suppresses the slow carrier yet *passes* this line, so its inter-qubit contrast survives the pulse sequence instead of being decoupled away. That is what turns the cross-spectrum from a free-induction curiosity into a decisive, decoupling-robust handle on entanglement storage (Sec. 3).

A line on the exchange channel only. An independent coupler-defect process (TLF resonance at $\omega\tau = 0.236$ with a weak Connors-type knee) enters ζ_{12} alone. Single-qubit QNS cannot see it; its structure is reserved for the two-qubit spectra, and it is what the idle gate’s ZZ -decoupling has to dodge.

A quiet — but measurable — electrical background. A $1/f^{0.9}$ floor (the measured charge-noise exponent class) roughly 25 dB below the anchored natural-Si level. This is the contrast knob, and it is deliberately placed *at* the reconstruction’s sensitivity rather than below it: between the fourth harmonic and the 0.372 line it leaves the quiet band $\omega\tau \in [0.26, 0.31]$ whose *measured* level, not any line, sets what an informed design can reach.

The scenario parameter. A minimum pulse separation of 8τ (40 ns π -pulse spacing at the anchor) applies identically to the CDD library, the NT optimizer, and every counterfactual. Without it, separation-limited trains park their passbands above the comb’s measured band and the comparison decouples from the reconstruction; with it, every sequence lives on spectra the protocol has measured.

quantity	value	anchor
time unit (min. pulse separation / 8)	$\tau = 5$ ns	Ge-hole / fast-Rabi anchor
T_2^* (both qubits)	$3500\tau = 17.6\mu\text{s}$	measured Ge-hole T_2^*
Gaussian FID $T_2 = 2/S_{a,a}(0)$	$3.10 \times 10^3\tau$	model output
slow-carrier correlation time / share	$400\tau / c_{\text{QS}} = 0.95$ common-mode	Yoneda low- f coherence class
local defect line + harmonics	$\omega_0\tau = 0.051, \{2, 3, 4\}\omega_0$	^{73}Ge -line / trap family
shared-TLF line (common-mode)	$\omega\tau = 0.372$; on $S_{1,1}, S_{2,2}, \text{Re } S_{1,2}$	correlated-defect class
coupler (ZZ) line + knee	$\omega\tau = 0.236$; knee at 0.025	independent coupler TLF
electrical floor	$1/f^{0.9}$, -25 dB vs anchored	charge-noise exponent class
quiet band (between lines)	$\omega\tau \in [0.26, 0.31]$	set by the engineered lines
control bandwidth	min. pulse separation $8\tau = 40$ ns	scenario parameter
gate windows studied	CZ: $T_G = 40\text{--}2560\tau$; idle: $320\text{--}10240\tau$	$0.01\text{--}3.3T_2^*$

Table 1: The dephasing scale and where the gates sit relative to it.

2 Characterization: the blind reconstruction recovers the truth

The protocol is identical to the anchored study — the same comb, the same estimators, the same DC experiments — run once at the budget a one-time device characterization affords: 128k shots per estimator, with the harmonic sweeps extended to 2560τ ($0.73T_2^*$; about fifteen minutes of simulation wall time). The acceptance standard is the one set in the June 12 discussion: *within error bars everywhere, with bars that inform*. Table 2 gives the result against the analytic truth, per spectrum.

Two honest footnotes to “tight everywhere”. First, harmonics adjacent to the lines on the smallest channels carry locally widened statistical bars: the probe sequences decohere faster while sweeping across a line, a measurement back-action that no shot budget removes; those teeth remain within bars. Second, the slope-fit DC experiments on $S_{1,1}/S_{2,2}$ cannot see the full quasistatic carrier and are flagged *undetermined* by the protocol itself (the stored values are first-harmonic floors); the double-echo $S_{12,12}(0)$ experiment, by contrast, is determined and lands on truth to better than 1%. The flags propagate to the gate stage, which treats those DC points accordingly — part of the protocol, not a patch.

SPAM. The three-arm comparison (Fig. 2) reruns the full experiment suite with hardware-realistic SPAM injected ($\alpha_M \sim 0.95\text{--}0.97$, $\alpha_{\text{SP}}^z \sim 0.98\text{--}0.99$, nonzero readout asymmetries and transverse preparation errors),

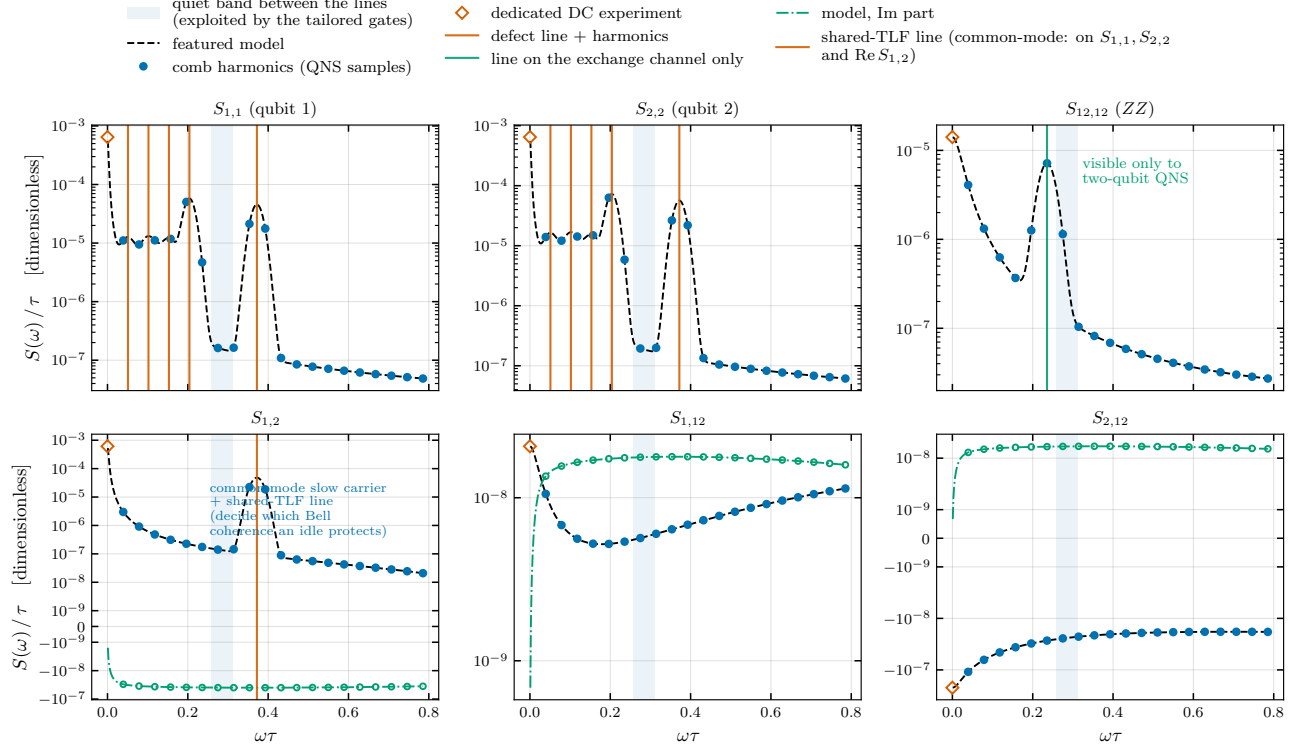


Figure 1: The featured two-qubit dephasing model. Each panel is one of the six spectra the QNS protocol reconstructs: dashed black is the analytic model (top row: the two single-qubit channels and the ZZ channel; bottom row: the three cross-spectra, real parts in black, imaginary parts in green). Blue dots mark the 20 comb harmonics $\omega_k\tau = 2\pi k/160$ the protocol samples; open diamonds are the dedicated DC experiments. Vertical orange lines: the local defect harmonics on $S_{1,1}/S_{2,2}$, and — now also in the $S_{1,2}$ panel — the shared-TLF line at $\omega\tau = 0.372$, which being common-mode sits on the self-spectra *and* in $\text{Re } S_{1,2}$. The coupler resonance (green) lives on $S_{12,12}$ *only* — structure invisible to single-qubit QNS. In $\text{Re } S_{1,2}$ the common-mode slow carrier gives the low- f rise and the shared TLF the resonant peak; together they decide which Bell coherence an idle protects, and the peak — unlike the carrier — lies in the decoupling passband, so it survives the train. Shaded band: the quiet window between the lines that the tailored designs exploit.

spectrum	median rel. dev.	median pull	max pull	within 2σ
$S_{1,1}$	1.0%	0.32	1.79	100%
$S_{2,2}$	2.1%	0.20	2.05	95%
$S_{12,12}$	1.7%	1.00	2.36	95%
$S_{1,2}$	3.3%	0.24	1.84	100%
$S_{1,12}$	15.1%	0.64	2.15	95%
$S_{2,12}$	11.1%	0.84	1.88	100%

Table 2: Blind 128k reconstruction vs. truth (pull = deviation / quoted total uncertainty; 20 harmonics per channel). Every tooth of every channel sits within 2.4σ of truth — across the 120 teeth, the handful in the $2\text{--}3\sigma$ band is what correct coverage predicts. The bars inform: median signal-to-bar ratios run 4–50 per channel, and every engineered feature — the defect harmonics on $S_{1,1}/S_{2,2}$, the coupler line on $S_{12,12}$, and now the shared-TLF line resolved in $\text{Re } S_{1,2}$ at $\text{SNR} \approx 15$ — is detected at 5σ or better. The double-echo DC experiment lands $S_{12,12}(0) = 1.41 \times 10^{-5}$ against a true 1.41×10^{-5} . The two exchange-cross spectra $S_{1,12}, S_{2,12}$ stay the floor-level channels (median S/N $\sim 4\text{--}6$, deviations 11–15% but every tooth within bars): their only structure is the shared electrical fraction, which no common-mode feature reaches.

at 64k shots per arm on *identical noise* (record/replay), so the arms differ only in what SPAM does and what

mitigation recovers. The result reproduces the anchored study’s conclusion on the featured landscape: the mitigated arm is statistically indistinguishable from the SPAM-free reference (maximum deviation 0.02σ across all six spectra), while the raw arm is badly biased — and on this stronger landscape the bias is large in σ , not merely in relative terms, because the resolved features now carry tight bars: a median 17σ on the high-SNR ZZ channel (up to $\sim 150\sigma$ at its peak) and tens of σ on the self-spectra. In *relative* terms the corruption is $3\text{--}4\times$ (350–430%) on every self- and ZZ channel — the carrier head, the line heights, the ZZ floor — and larger still on the floor-level exchange crosses. The SPAM-robust protocol remains immune by construction at its stated price (it cannot reconstruct $S_{1,12}, S_{2,12}$); whether that price matters is a gate-level question, answered next.

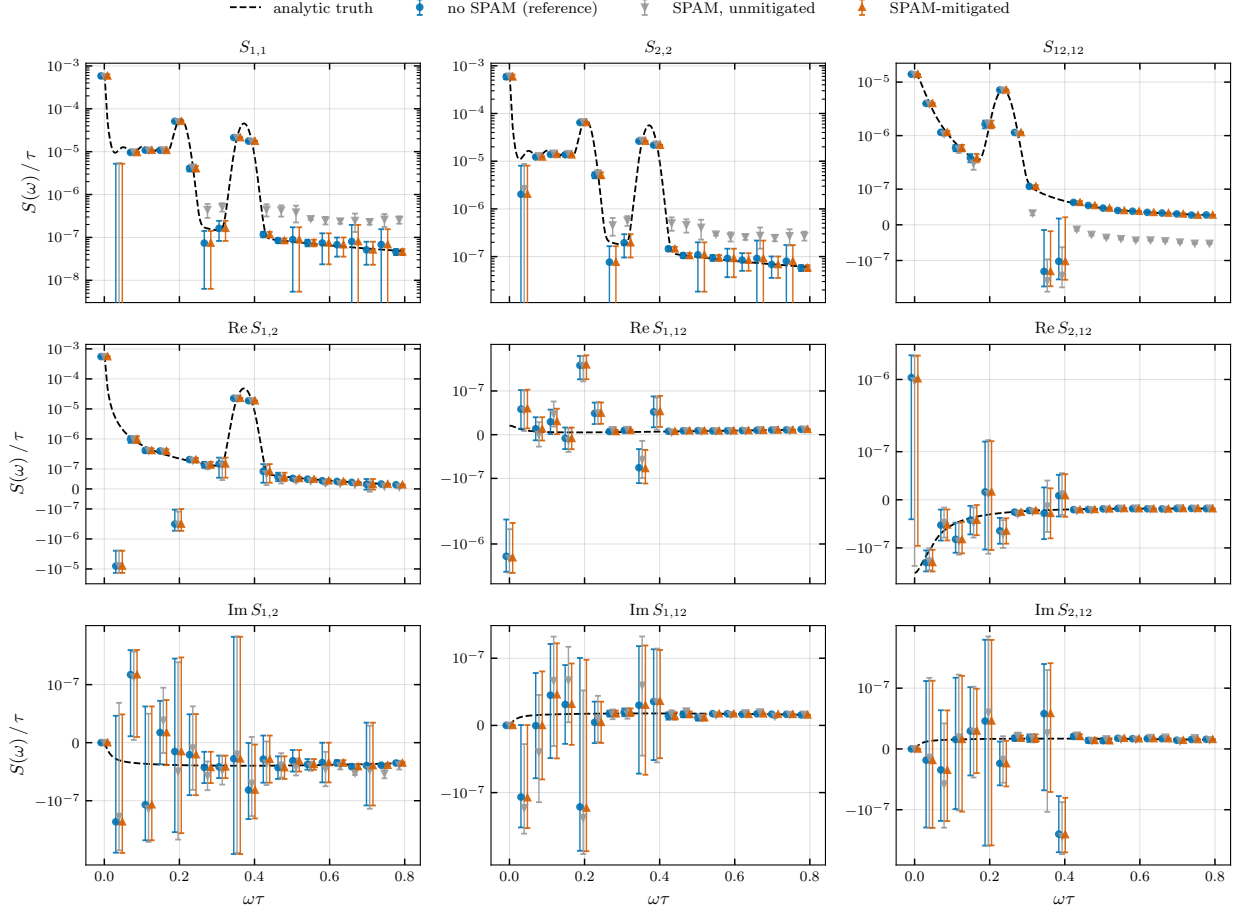


Figure 2: The three reconstruction arms at 64k shots on identical noise, against analytic truth (dashed): reference (no SPAM, blue circles), raw (SPAM, unmitigated, grey triangles), SPAM-mitigated (orange). Bars are the quoted total (statistical + systematic) uncertainties; arms are slightly offset in ω for legibility. Top row: the three self-spectra; middle/bottom rows: real and imaginary parts of the cross-spectra. The raw arm’s bias is largest on the ZZ channel and the line heights; the mitigated arm sits on the reference everywhere.

3 What the gate design needs from the spectra, and what the physics does

With a reconstruction that captures the landscape, the question is how much of it the control design actually consumes — and what the parts the design does *not* consume are for. We answer with the same two experiments as the anchored study, run for both gates (Fig. 3), plus the demonstration the cross-spectra exist for (Fig. 4).

Two-qubit QNS is necessary to design the entangler. Re-running the blind NT optimization with the optimizer’s channels progressively restored — the two single-qubit spectra alone $\rightarrow$ the three self-spectra $\rightarrow$ the

SPAM-robust four $\rightarrow$ all six — the entangling gate steps once, by $3.1\times$, and only at the ZZ rung: 2.28×10^{-3} from the single-qubit spectra alone versus 7.31×10^{-4} once $S_{12,12}$ is supplied, on the true noise at $T_G = 320\tau$ (Fig. 3a). Blind to the ZZ -correlated dephasing the gate accumulates, the single-qubit design drives its own self-only objective to $\sim 5 \times 10^{-4}$ while sitting at 2.28×10^{-3} in truth; given $S_{12,12}$ the optimizer shapes the y_{12} filter against it. That a two-qubit channel can decide a high-fidelity design is fixed by how the average fidelity reads the noise: to leading order in a dephasing model,

$$1 - F_{\text{pro}} = \sum_a w_a \chi_{aa} + O(\chi^2), \quad \chi_{ab} = \int \frac{d\omega}{2\pi} \mathcal{F}_{ab}(\omega) S_{ab}(\omega), \quad (1)$$

where $a, b \in \{1, 2, 12\}$ index the $Z_1, Z_2, Z_1 Z_2$ dephasing generators and $w_a > 0$. The three *self*-spectra ($S_{1,1}, S_{2,2}, S_{12,12}$) enter at *first* order; every *cross*-spectrum cancels in the Pauli trace that defines F_{pro} — $\sum_P s_P^a s_P^b = 0$ for $a \neq b$, the $\pm$ filter combinations pairing off — and survives only at *second* order, $O(\chi^2)$. A high-fidelity design is therefore set by whichever self-spectrum the gate cannot escape, here $S_{12,12}$; the cross-spectra (robust-4, all six) add nothing, so the CZ ladder is the single step the first-order argument demands. The idle shows the same step, smaller ($1.19\times$ at 640τ , $1.54\times$ at 10240τ , best over the repetition number M). $S_{12,12}$ — the ZZ -correlated dephasing — is reconstructible only by the two-qubit protocol, and this design gap is its operational necessity, not a characterization nicety.

SPAM-biased spectra. Handing the optimizer the raw arm’s biased reconstruction yields the *identical winner at the identical true infidelity* for both gates — the same NT(27,30) entangling design at 7.31×10^{-4} and the same idle winner at 1.57×10^{-3} as the reference and mitigated arms (Fig. 3, right column): the *true* design is robust because the raw arm’s corruption sits on frequencies the *winning* filters barely touch. The *prediction*, though, is not — and here, unlike the carrier-only landscape where the raw arm happened to predict correctly, it pays for the bias: the raw arm predicts 1.11×10^{-3} for the CZ where the reference and mitigated arms predict 1.03×10^{-3} (an 8% over-statement), and 2.11×10^{-3} vs 1.95×10^{-3} for the idle. The reference and mitigated predictions agree to 0.1%; mitigation removes the bias. A device map drawn from raw data mis-states the self- and ZZ -channel levels by $3\text{--}4\times$, and mitigation repairs all of it for two orders of magnitude less than the cost of being wrong about a trap’s depth.

What full, unbiased characterization buys is everything around the design:

- **The error budget.** Under the best decoupling the two-qubit channels ($S_{12,12}$ plus cross-spectra) carry **45%** of the entangling gate’s residual at $T_G = 320\tau$ and **22.5%** of the deep idle’s (2560τ): nearly half of the tailored CZ’s error is ZZ -correlated dephasing a single-qubit campaign cannot see. With the design itself $3.1\times$ worse without $S_{12,12}$ (above), two-qubit QNS is necessary both to *build* a decoupled entangler and to *certify* its error budget.
- **The prediction.** The reference-arm predictions are uniformly conservative ($\sim 1.3\text{--}1.4\times$ above truth: the reconstruction reads the quiet band slightly high and treats the flagged DC points pessimistically), and *certified*: every uncertainty band in Sec. 4 exists only because the bars in Table 2 are unbiased with stated coverage. With biased central values (the raw arm above) the same bands would be confidently wrong.
- **The spectroscopy itself.** The raw arm’s spectra are relatively corrupted by $3\text{--}4\times$ on the self- and ZZ channels that set every trap’s depth and the ZZ floor — in σ , a median 17σ on the ZZ channel and up to $\sim 150\sigma$; mitigation recovers the reference to 0.02σ .

What the cross-spectra are for. The off-diagonal spectra ($S_{1,2}, S_{1,12}, S_{2,12}$) barely move either ladder above — and that is a theorem, not a landscape accident: in every PTM element the cross-terms enter through $\pm$ filter combinations whose first-order contributions cancel pairwise in the *average* fidelity, surviving only at $O(\chi^2)$. Average fidelity is the wrong meter for them. Their meter is element-resolved: *which* coherence survives. The idle’s defining task makes this concrete (Fig. 4): hold half of a Bell pair, $\Phi^+ = (|00\rangle + |11\rangle)/\sqrt{2}$, through the idle. The Bell coherences are exactly ZZ -immune ($|00\rangle$ and $|11\rangle$ share the $Z_1 Z_2$ eigenvalue), so only $S_{1,1}, S_{2,2}$ and $\text{Re } S_{1,2}$ enter: Φ^+ decays through the $S_{1,1} + S_{2,2} + 2\text{Re } S_{1,2}$ combination under in-phase decoupling ($y_2 = +y_1$: simultaneous pulses, the hardware default) and through $S_{1,1} + S_{2,2} - 2\text{Re } S_{1,2}$ under anti-phase decoupling ($y_2 = -y_1$: the same train with one qubit’s idle bracketed by frame flips). Both common-mode sources — the slow carrier and the shared TLF — live in $\text{Re } S_{1,2}$, so anti-phase places the pair in a decoherence-free subspace with respect to *both*, while in-phase is maximally exposed to both.

In *free* evolution the carrier dominates and the split is the bare $(1+c)/(1-c)$ physics: Ψ^+ outlives Φ^+ by $33\times$ at 320τ (1.81×10^{-3} vs 5.93×10^{-2}), a structure no single-qubit measurement can reveal. But free induction

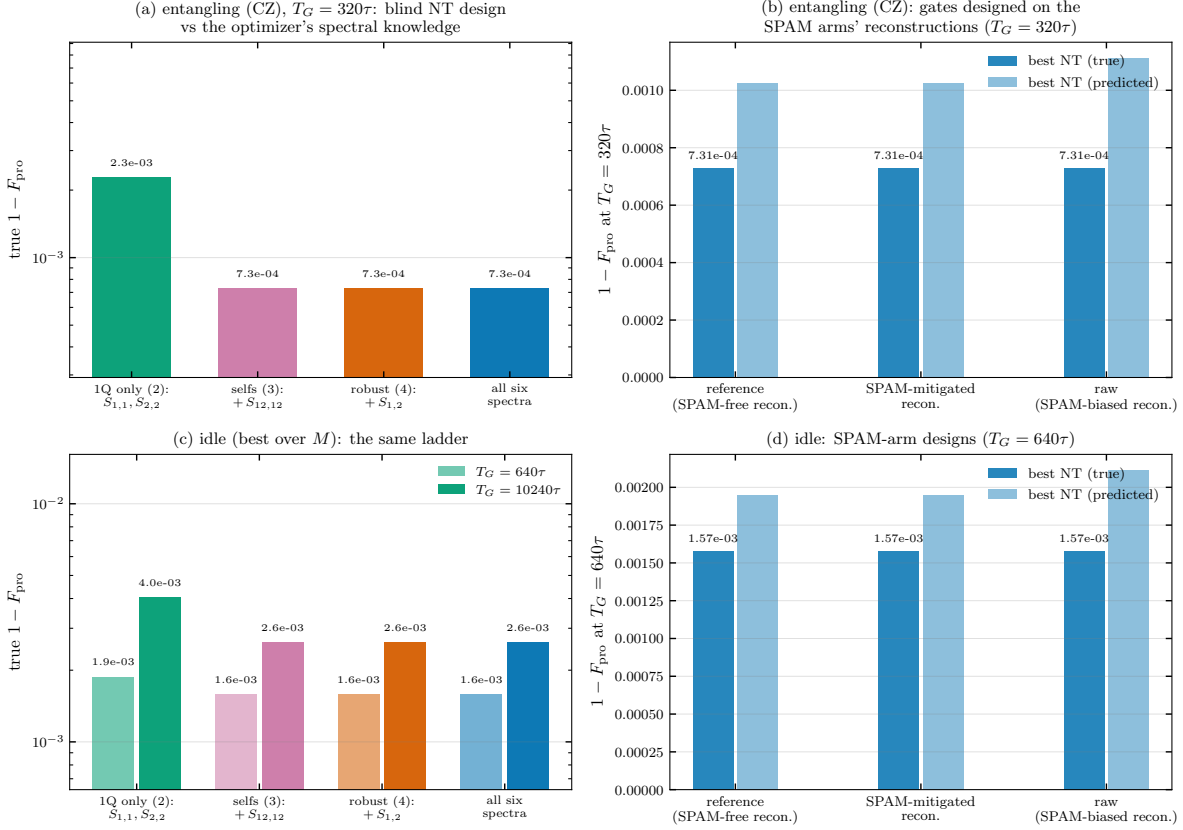


Figure 3: Both experiments for both gates: entangling (top row) and idle (bottom row, best over M ; light/dark bars: $T_G = 640\tau$ and 10240τ). Left column: true infidelity of the blind NT winner as the optimizer's spectral knowledge grows from single-qubit-only, through the three selfs and the robust protocol's four, to all six spectra. Both ladders step once and only at the $S_{12,12}$ rung — the ZZ -correlated dephasing a single-qubit campaign cannot see ($3.1\times$ for the CZ, 1.2 – $1.5\times$ for the idle) — and are flat across the cross-spectra rungs, which average fidelity does not consume. Right column: gates designed on the SPAM arms' reconstructions; true infidelity (solid) and the designer's own prediction (light). The raw arm's bias surfaces in the prediction, not the design.

is not how one stores a pair — one decouples, and that is exactly where the carrier-only landscape was weak: a ZZ -decoupling train *suppresses* the carrier, the very correlation the split rode on, collapsing the decoupled advantage to 1.2 – $3.3\times$. The shared TLF repairs precisely this. Its correlation sits at $\omega\tau = 0.372$, *inside* the train's passband, so decoupling passes it and the anti-phase DFS cancellation now acts on a line-scale feature the train cannot remove. The two implementations of the *same pulse train* — identical single-qubit marginals, average fidelities equal to the quoted precision (computed ratio 1.000 at every gate time), indistinguishable to any single-qubit QNS campaign — now store the pair a **robust** 8 – $10\times$ **apart** across the entire idle window ($9.8\times$ at 320τ through $8.1\times$ at 10240τ : 2.9×10^{-4} vs 2.9×10^{-5} , rising to 1.6×10^{-3} vs 2.0×10^{-4}). The *plateau* is the point. Where the free-induction split decays with storage time as the carrier saturates Φ^+ ($33\times$ at 320τ falling to $3.6\times$ at 10240τ), the decoupled split holds flat and past $\sim 5000\tau$ *exceeds* it — the shared line keeps delivering contrast through the train long after free evolution has stopped distinguishing the parities.

Which implementation protects the pair is decided by the *sign* of the measured $\text{Re } S_{1,2}$ (here > 0 , so anti-phase wins; on a device with anti-correlated structure the assignment flips); the magnitude — the shared-TLF peak, reconstructed at $\text{SNR} \approx 15$ — prices the payoff. The blind reconstruction picks the right parity at every gate time and predicts a split of the same order (~ 12 – $16\times$, conservatively over-stating the in-phase penalty as it reads the quiet band slightly high); a record/replay Monte Carlo through the synthesized noise reproduces the formula values to 0.6 – 1.3% . The blind average-fidelity NT winner sits between the two parities — correctly, since average fidelity cannot see the choice.

This is the idle's version of the entangler's story, told channel for channel. Two-qubit QNS is necessary to *design*

the CZ — the $S_{12,12}$ rung, $3.1\times$ — and it is equally necessary to *operate* the idle: blind to the two-qubit cross-spectrum one runs the hardware-default in-phase train and forfeits the full $\sim 9\times$. The CZ’s necessity surfaces in average fidelity through a self-spectrum; the idle’s surfaces element-resolved through a cross-spectrum; both are invisible to a single-qubit campaign, and on this landscape both are decisive.

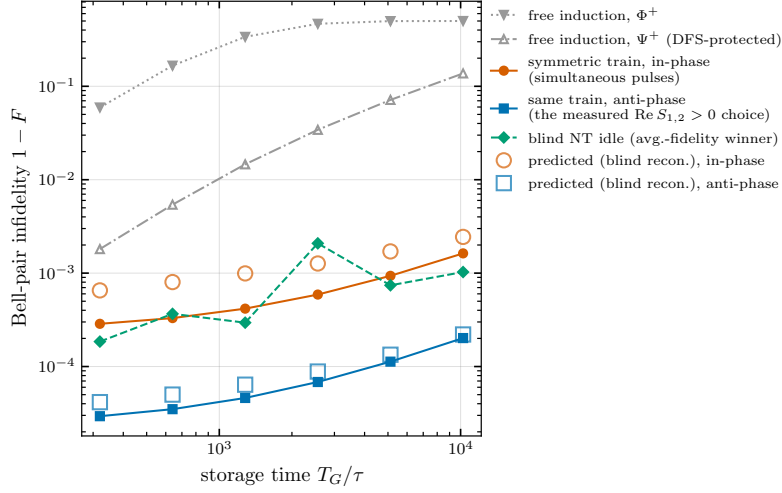


Figure 4: Holding a Bell pair through the idle. Free induction realizes the bare $(1 + c)/(1 - c)$ split between Φ^+ (grey, filled) and Ψ^+ (grey, open): $33\times$ at 320τ , but *decaying* as the carrier saturates Φ^+ (to $3.6\times$ at 10240τ) — correlated-noise physics invisible to single-qubit data. Under decoupling the same symmetric train, implemented in-phase (simultaneous pulses, vermillion) and anti-phase (frame-flipped, blue) — implementations with identical average fidelity — splits by a *robust* $8\text{--}10\times$ across the whole window, because the shared-TLF correlation at $\omega\tau = 0.372$ lies in the train’s passband and survives it (where the DC carrier does not). The decoupled split holds flat and overtakes the free-induction split past $\sim 5000\tau$. Open markers are the values predicted from the blind reconstruction, which picks the right parity everywhere and the magnitude to the same order. The blind average-fidelity NT idle (green) sits between the parities, as it must.

4 The gates: margins for both operations

Figure 5 summarizes the gate results on the 128k reconstruction, all under the 8τ pulse-spacing scenario and the blind protocol: select on the characterized spectra, report the true infidelity. Both June 12 requirements hold at once. *Realistic infidelities*: the tailored entangling gate holds in the $10^{-4}\text{--}10^{-3}$ band across $T_G = 40\text{--}2560\tau$ — 7.3×10^{-4} at 320τ , rising to 6.2×10^{-3} at 2560τ as the forced ZZ exposure accumulates — while tailored idling holds at the low- 10^{-3} level out to $10240\tau \approx 3T_2^*$. *A robust margin over the best library sequence*: the entangling margin climbs from $1.5\times$ at 40τ to a $\sim 10\times$ plateau across $320\text{--}2560\tau$ (9.8, 11.4, 9.7, $9.4\times$); the plateau — rather than the unbounded growth of a ZZ-free landscape — is itself the signature of the forced ZZ residual no CZ design can dodge. The idle margin grows from $\sim 2\times$ at short times to $\sim 24\times$ at 10240τ , for the structural reason of Sec. 1: each doubling of the gate time slides the CDD library’s passbands onto the next defect-line harmonic, while the tailored design keeps using the measured quiet band.

The margins, with their 95% confidence intervals under reconstruction uncertainty:

- **Entangling (CZ):** $1.5\times [1.78, 1.81]$ at 40τ ; $2.2\times [1.66, 1.71]$ at 80τ ; $5.9\times [4.14, 4.20]$ at 160τ ; $9.8\times [6.79, 6.92]$ at 320τ ; $11.4\times [9.46, 9.62]$ at 640τ ; $9.7\times [9.47, 9.59]$ at 1280τ ; and $9.4\times [9.58, 9.69]$ at $2560\tau \approx 0.8T_2^*$ — a plateau, not the unbounded climb of a ZZ-free landscape: the forced ZZ residual caps the tailored gate at depth as well.
- **Idle:** $2.1\times [2.37, 2.53]$ at 320τ ; $2.2\times [2.12, 2.27]$ at 640τ ; $2.9\times [2.73, 3.01]$ at 1280τ ; $2.4\times [1.97, 2.31]$ at 2560τ ; $14.2\times [15.4, 19.9]$ at 5120τ ; and $24.0\times [24.8, 37.8]$ at $10240\tau \approx 3T_2^*$, once the deep idle’s repetition count lets it dodge the ZZ knee.

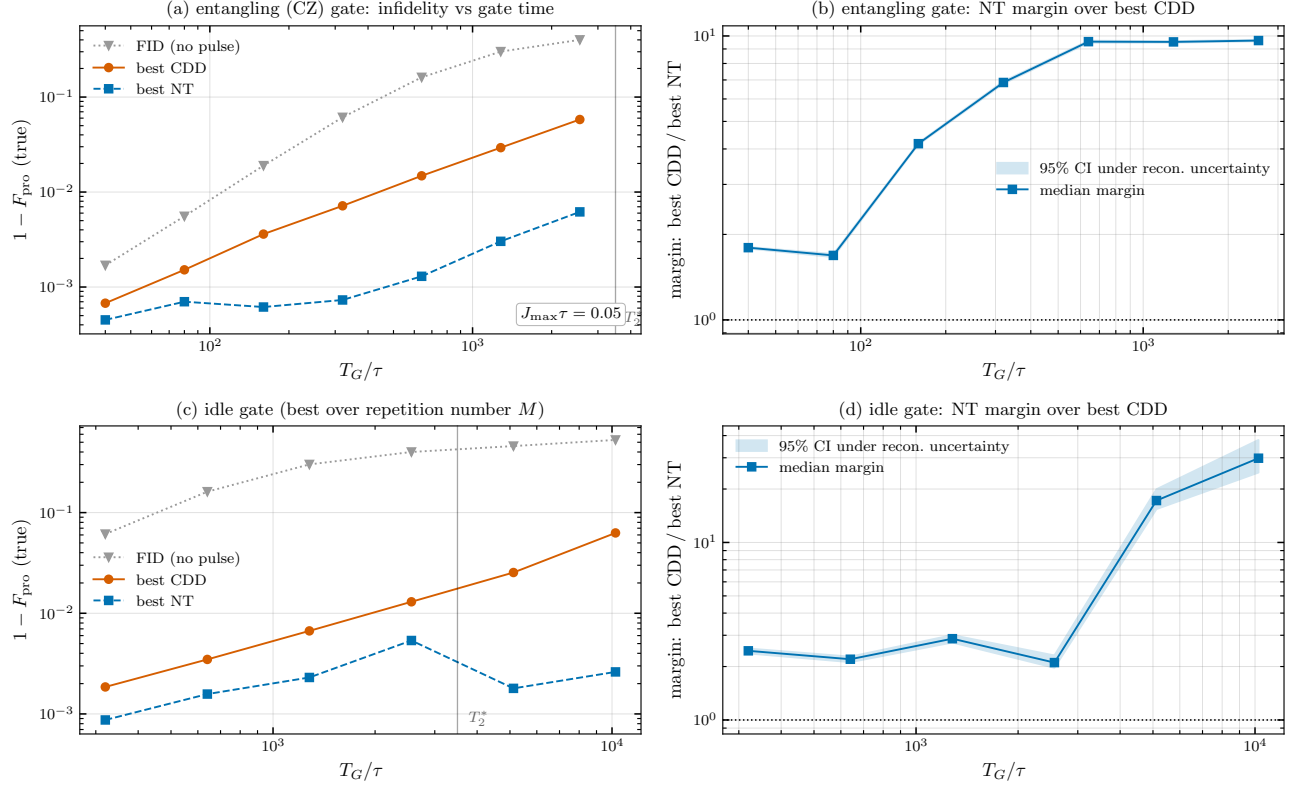


Figure 5: Noise-tailored gates on the featured model, all design blind, all scoring on truth, pulse-spacing constraint 8τ throughout. (a,c): true process infidelity vs. gate time for the entangling and idle gates: free induction (grey), best CDD (blind-selected, orange), best NT (blind-optimized, blue); the vertical line marks $T_2^* = 3500\tau$ (the idle is optimized over the repetition number $M \leq 128$ at each gate time). (b,d): the corresponding NT-over-CDD margin with the 95% band obtained by propagating the reconstruction’s quoted statistical and systematic uncertainties (systematics drawn coherently per channel, tail fits redone per draw) through both fixed winner sequences; the band is on the *ratio*, so common-spectrum fluctuations cancel.

In each entry the central value is the *true* margin and the bracket is the 95% interval of the *predicted* margin under reconstruction uncertainty. From 160τ on, the bands track the true margins and sit conservatively below them: the reconstruction’s slightly-above-truth reading of the quiet band makes every predicted NT infidelity an upper estimate — the pipeline does not flatter its own gates. At the two shortest gate times, where best CDD and best NT sit within $2\text{--}3\times$ of each other at the few- 10^{-4} floor, the predicted ratio is least accurate (true $1.5 \times / 2.2\times$ vs. predicted $\sim 1.8\times$); the quoted true values are the honest ones there.

operation	FID	best CDD	best NT	margin
idling 640τ	$1.67\text{e-}1$	$3.48\text{e-}3$	$1.57\text{e-}3$	$2.2\times$
idling 10240τ	$5.28\text{e-}1$	$6.28\text{e-}2$	$2.62\text{e-}3$	$24.0\times$
entangling 320τ	$6.12\text{e-}2$	$7.16\text{e-}3$	$7.31\text{e-}4$	$9.8\times$

Table 3: Sweet-spot numbers on the featured model (true process infidelities, blind designs, 8τ pulse-spacing scenario).

Where this leaves the two-scenario story: the anchored model says that on a quiet, monotonic, well-behaved device, generic DD is nearly optimal and tailoring buys a reproducible $1.5\text{--}1.7\times$. The featured model says that the moment a device carries resolvable structure — defect harmonics, coupler resonances, common-mode drift, the things purification demonstrably does not remove — the same characterization pipeline, run once at a budget a real experiment can afford, separates 10^{-2} -class from $10^{-4}\text{--}10^{-3}$ -class two-qubit gates; that the two-qubit

reconstruction is operationally necessary for *both* gates, symmetrically — to *design* the entangler (the $S_{12,12}$ rung, a $3.1\times$ design step in average fidelity) and to *operate* the idle (the shared-TLF cross-spectrum $\text{Re } S_{1,2}$, a decoupling-robust $\sim 9\times$ in entanglement storage whose very sign a single-qubit campaign cannot measure); and that no shortcut (a single-qubit campaign, unmitigated SPAM) delivers the same gates with the same certified predictions. Which device one has is an empirical question — and answering it is itself the reconstruction.

All optimization blind; true infidelities evaluated exactly on the analytic model; characterization at 128k shots per estimator, SPAM arms at 64k on identical noise; pulse budgets separation-limited under the 8τ spacing scenario for both the NT search and the CDD library.